



MoE-AICTE
INVESTOR NETWORK

MoE-AICTE Investor Network





HARINARAYAN RECYCLERS LLP

Dr.Vivek Sharma
National Awardee

**Ministry of Chemicals &
Fertilizers Govt.of India,**

New Delhi

**National Entrepreneur
Award in the State wise
category by Entrepreneurs
Association of India**

- **Establishment Year & Location:**
2024,Bathinda151001-Punjab-India
LLP IN: ACG-6154 , DIPP 163735
GST No. 03AAQFH9844M1ZO

- **Incubation/Funded Institute:**
 - AIC Indian Institute of Science Education and Research (IISER),Pune.
 - Punjab state Council for Science Technology, Chandigarh.
 - Technology Enabling Centre (TEC),Panjab University
 - Entrepreneurs Association of India(EAI).



AIC IISER PUNE



Pitch Deck Outline

- 1) Elevator Pitch**
- 2) Problem & Current Solutions: What need do you fill? Other solutions**
- 3) Product or Service: Your solution**
- 4) Business Model: Key Revenue Streams**
- 5) Competition**
- 6) Market Opportunity: Define market size & your customer base**
- 7) Momentum, Traction, Expertise: Your key numbers**
- 8) Market Approach & Strategy: How you grow your business**
- 9) Financials**
- 10) Team & Key Stakeholders (Investors, Advisors)**
- 11) Investment: Your 'Ask' for funding, Basic use of funds**



Vision / Elevator Pitch

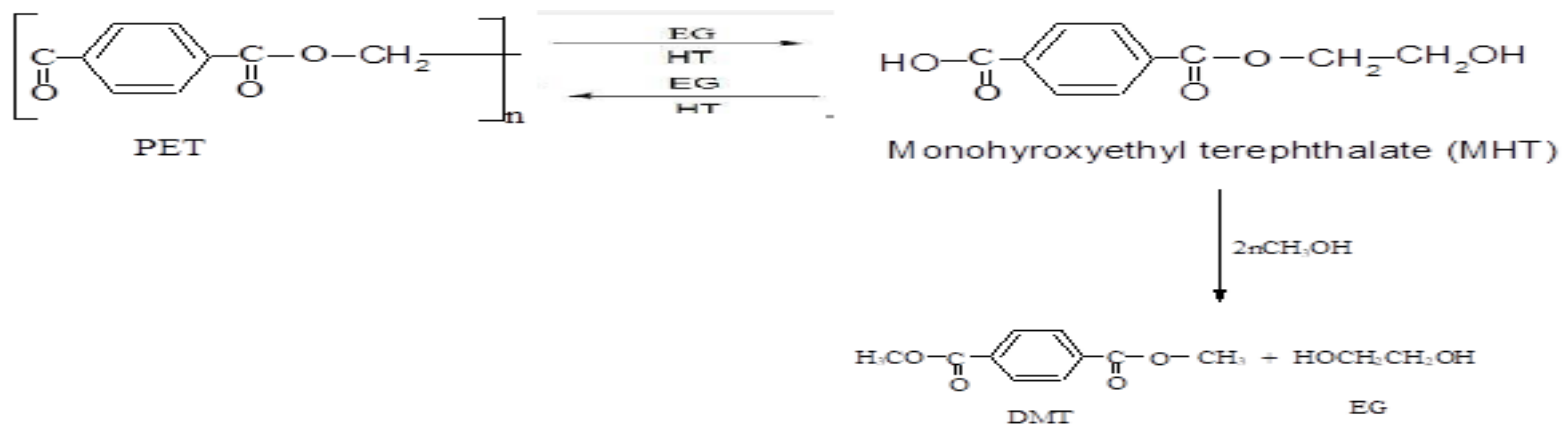
"We are the Green Revolution for Plastic, turning waste into valuable resources to create a cleaner, sustainable future."

The Problem

India would see a rise in PET waste generation from less than 40,000 metric tons per year to over 125000 metric tons by the year 2030. Only 20% plastic are recycled, remaining are landfilled, Burned and ends up in water bodies so it causes hazardous effects on all living organisms. Today, PET-Waste have become one of the most valuable and successfully recyclable materials .

Current Solutions

Recycling Reaction



Burning issues



Landfill issues

Products



Pure PET



DMT



EG

Proposed Idea Stage:

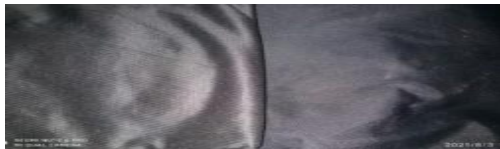
Developed a prototype at the kilo scale and Granted **Indian Patents No. 446431 (22/08/2023) and 486052 (20/12/2023).**

Next step: scaling up for commercial production.

The Solution- Product / Service

Raw Materials

NIKE PET WASTE



Waste PET Bottles



Products

Pure DMT

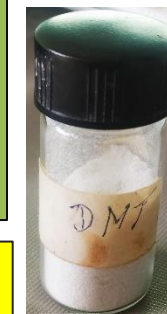
Pure PET

Product Features

- Highly pure monomer DMT & EG
- Inexpensive Renewable catalysts and solvent system
- Zero Micro content Dissolution

- Chemical Resistance
- Pure PET can be used as food grade applications
- Light weight and Environment friendly.

Products



DMT

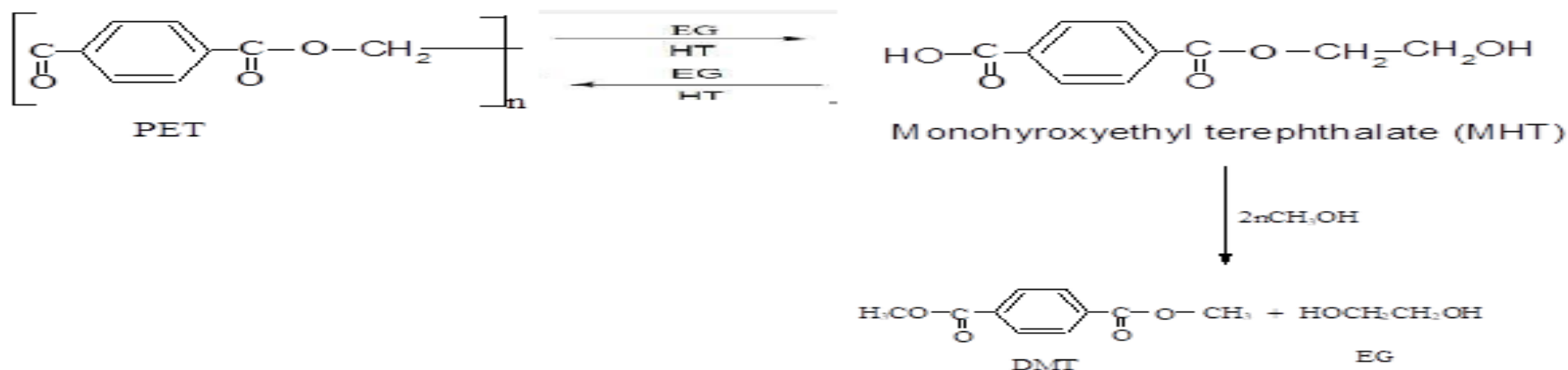


EG



Pure PET

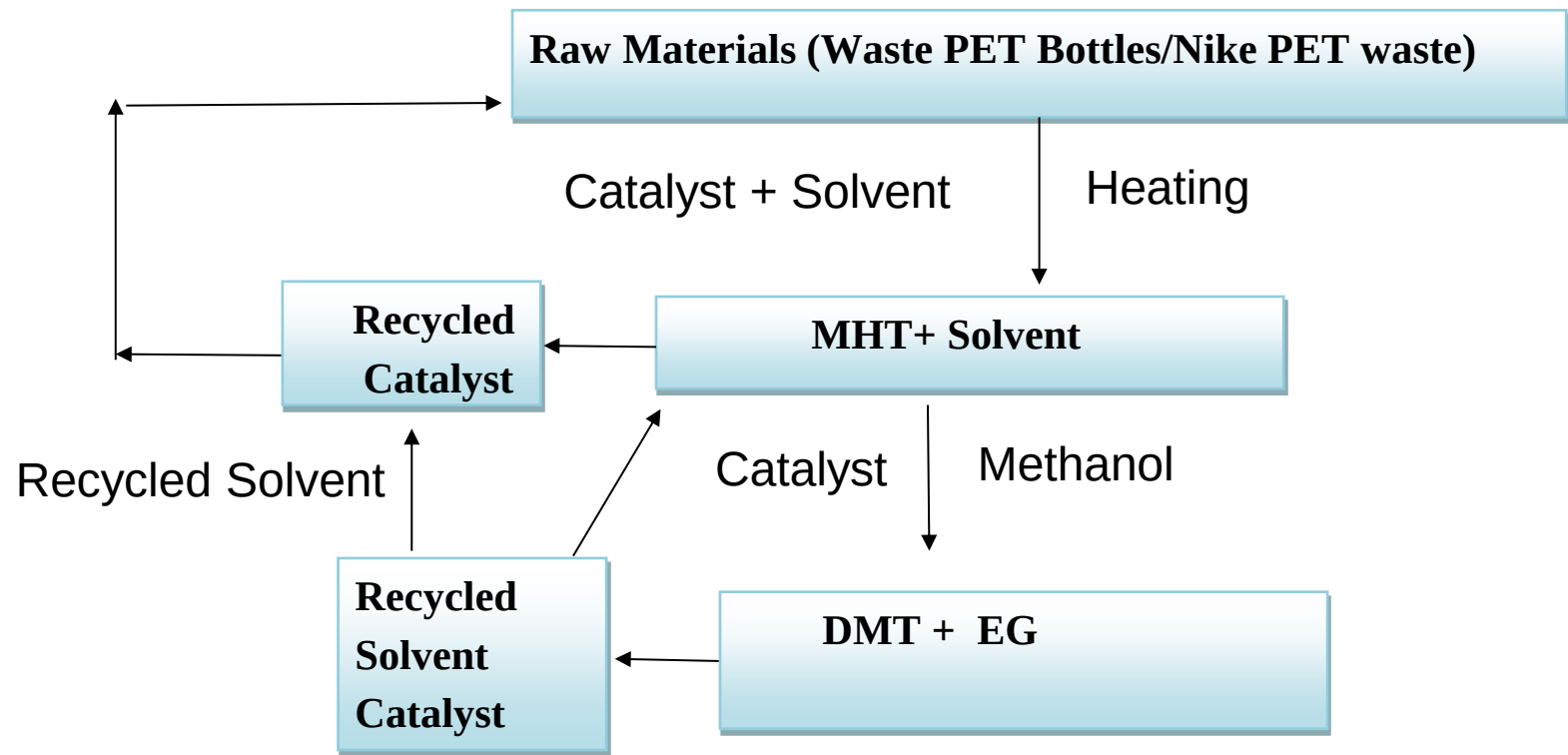
Recycling Reaction





Ease Of Manufacturing

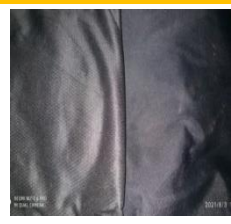
Recycling of waste plastic bottle to high grade PET and Dimethyl Terephthalate (DMT) and Ethylene Glycol (EG) 'A Dual Application Approach'



Raw Materials



PET Waste Bottle



NIKE PET Waste

Products



DMT



EG



Pure PET

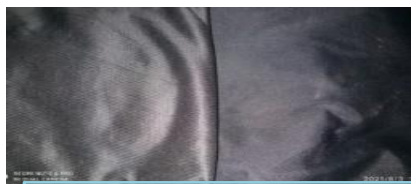
Revenue Model

1000kg Batch Expected Cost

Costs Breakdown

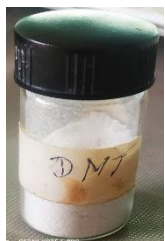


Waste PET Bottles



Nike PET Waste

Revenue Breakdown



DMT



EG



High
Grade
PET

Costs Breakdown	Expected Price (Rs.)	
Raw Materials (Clean PET Chips)	60,000 (Variable as per market price)	
Processing Cost	50,000	
Total Cost	110,000	
Product	Expected Price (Rs.)	GC-MS Purity of obtained Product
DMT Revenue (Rs.185/Kg) 1010kg	1,86,850	99%
EG Revenue (Rs. ₹80/Kg) 322.9kg	25,832	99%
Total Revenue	212,682	
Per Batch Profit	102,682	

Primary Customers: Industries in textiles, packaging, and plastics, including NIKE PET Textile Industry, Shivam Plastic, Ludhiana, and others.

Basic Math on Revenues & Conversion Rates Production

Cost per Batch:

₹110,000

Profit Margin: ~53% Conversion

Rate: 100% of raw PET waste converted into high- grade products.

Key Highlights: High product purity (99%), strong profitability per batch, and established customer relationships ensure long-term growth and sustainability.

Competition

Market Space:

We specialize in the sustainable recycling industry, converting waste PET into high-grade PET and eco-friendly outputs like DMT and Ethylene Glycol.

Advantages:

Patented technology ensures 99% purity with eco-friendly, cost-efficient processes.

Dual revenue streams: product sales and waste management services.

Competitive pricing due to lower production costs using renewable catalysts.

Unique Market Position:

By combining sustainability, affordability, and superior quality, we are the preferred choice for industries seeking eco-friendly solutions. Strong local partnerships ensure a reliable supply chain and scalability.

Competitors:

Large-scale recyclers prioritize mass production but lack sustainable methods.

Smaller recyclers cater to niches but face scalability and innovation challenges.

Differentiation:

National recognition, including the **11th National Petrochemical Award**.

Advanced lab testing (GC-MS, NMR) guarantees superior product quality.

A unique focus on Tier-3 markets with a community-driven approach, fostering employment and sustainability awareness.

This strategy ensures a competitive edge, long-term growth, and loyal customers.

Market Opportunity

Market Definition:

We operate in the **sustainable recycling industry**, focusing on converting waste PET into high-grade PET and eco-friendly products like DMT and Ethylene Glycol.

Total Market Size:

The global recycled PET market is projected to reach **\$20 billion by 2027**, with a CAGR of 7%. India's recycling market represents a significant growth opportunity, driven by increased environmental regulations and corporate sustainability goals.

Niche Market:

Our focus lies in high-purity recycled PET catering to industries in **plastics, textiles, pharma, and food sectors**, and waste management services for institutions, municipal bodies, and event venues.

Target Customers:

Industrial Clients: Require high-grade PET for manufacturing.

Waste Generators: Institutions, municipal corporations, and event venues.

Macro Trends & Insights:

Increasing demand for sustainable materials due to government policies and ESG mandates.

Growing awareness among corporations and consumers about reducing plastic waste.

Rising raw material costs make recycled PET a cost-effective alternative.

This creates a vast opportunity for our innovative and sustainable recycling solutions to thrive in an expanding market.

Traction

Road-map to Commercialization

Projected Time to Hit the Market

- 1 year



Hurdles & Solutions

- Finance required to start production, applying for grants

*Funding: Rs. 6.5 L secured from AIC- IISER Pune and PSCST

Go to Market Strategy (GTM)

Market Validation: Engaged with 50+ stakeholders across textiles, pharma, cosmetics, and recycling industries; received strong demand for high-grade PET , DMT and EG.

Customer Access: Reach customers via partnerships with industry leaders, institutions, and targeted outreach in recycling and manufacturing sectors.

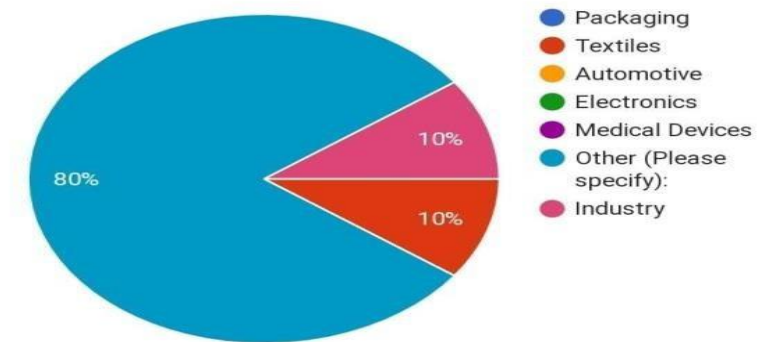
Growth Drivers: Leverage surveys, direct engagement, and sustainable branding to build credibility and capture market share.

Unique Edge: Patented chemical recycling method ensures efficiency, sustainability, and superior product quality, setting us apart from competitors.

Industry Sector (Select all that apply):

10 responses

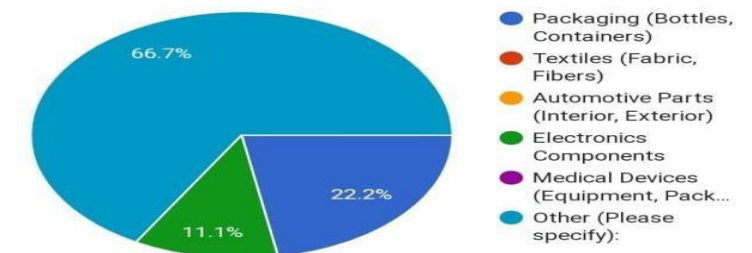
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Potential Applications of High-Grade PET in Your Industry:

9 responses

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Technology Enabling Centre (TEC)
(Sponsored by Deptt. of Science and Technology, Govt. of India)
PANJAB UNIVERSITY, SECTOR-14, CHANDIGARH-160014 (INDIA)
(Estt. Under the Panjab University Act VII 1947 enacted by the Govt. of India)

Prof. Manu Sharma
Coordinator, DST-TEC,
Hony. Director, CIIPP,
Professor (Mechanical Engg.), UIET,
Panjab University, Chandigarh.

Ref. No. : 425/2024-25/TEC/PU

Dated: 14-01-2025

To,

Dr. Vivek Sharma
Baba Farid College
Bathinda , Punjab

The Technology Enabling Centre, Panjab University Chandigarh in collaboration with Punjab State Council for Science and Technology, facilitated industry-academia linkage for Dr. Vivek Sharma's PET Recycling technology to support technology transfer. Dr. Virender Singh, Manager at DST-TEC facilitated ongoing discussions with the following industries are in progress aims to transfer technology to Industry and secure consultancy projects.

Sr. No.	Industry Name
1	Pyrotask Energy, Raigad, Maharashtra
2	The Shakti Plastic Industries, Mumbai
3	LUCKY PETROCHEMICALS
4	Meera Brijmohan Industries [O.P.C] Pvt Ltd, Maharashtra
5	Birla Textile Mills (A unit of Suttlej) Pvt Ltd, Baddi
6	The Kabadi Wala Startup, MP
7	Pashupati group Ltd, Uttarakhand
8	KKalpana, Kolkata

Regards

Dr. Ajay Sharma
Sr. Manager
Technology Enabling Centre
Panjab University, Chandigarh

Prof. Manu Sharma
Coordinator
DST-Sponsored TEC
Panjab University, Chandigarh

Financials

Revenue Model

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5-Year Commercialization Plan

Year 1: Establish a full-scale production facility in Bathinda.

Years 2-3: Scale production, diversify product lines, and secure IP for new technologies.

Years 4-5: Launch strategic marketing campaigns, form partnerships, and expand to new regions like Patiala.

Critical Assumptions

Expenses: Efficient raw material sourcing and energy usage.

Customer Conversion: Target 50+ stakeholders validated through surveys and Personal Interactions

Market Penetration: Gradual increase over 3 years, driven by demand for high-grade PET and DMT.

Yearly Financial Highlights

Year 1: Total valuation $\geq$ ₹10 Lakh, focus on infrastructure.

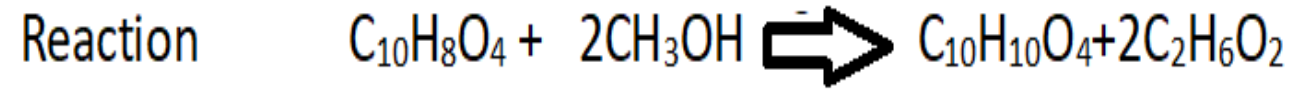
Year 2-3: Increase revenue by scaling operations and expanding market share.

Year 4-5: Achieve sustained profitability with diversified revenue streams.

EBITDA: Focus on consistent profitability with minimized operational costs.



Financials



Mol. Wt.	PET	Methanol	DMT	EG
(g/mol)	192	64	194	62

The excess weight in the final products is due to the **addition of methanol** during the reaction, which chemically combines with PET waste to produce DMT and ethylene glycol. The process is balanced, and the total input mass equals the total output mass.

Team



**PROJECT
LEADER**



**TEAM
MEMBER**



**SCIENTIFIC
ADVISOR**



**BUSINESS
ADVISOR**



**TEAM
MEMBER**

Role in Project	Dr. Vivek Sharma Technical Manager	Er. Ankita Pandey Marketing Manager	Prof. (Dr.) D.D. Agarwal Scientific Advisor	Ms. Iqbalpreet Kaur Sidhu Business Advisor	Dr. Heena Technical member
Designations	Assistant Professor	Assistant Professor	Professor	Incubation Manager	Assistant Professor
Affiliation	Baba Farid College ,Bathinda	Baba Farid College , Bathinda	Jiwaji University Gwalior (MP)	Baba Farid school of Entrepreneurship	Baba Farid College , Bathinda
Research Area	Organic Synthesis Polymer Sciences	Electronic & Communication	Homogeneous Catalysis Polymer Sciences	startup ecosystem	Inorganic Chemistry
Experience	12+ Years Teaching 6 Years Research	5+ Years Teaching 1 Years Industry	30+ Teaching & Research Experience	8 Years Experience in startup ecosystem	2 years Teaching Experience

INVESTMENT

Budget Heads	Amount (₹ 50 in lakhs)	Basis
Equipment	15 lakh	1,High pressure reactor is required for maintain pressure during reactions 2. Nitrogen determination (Kjeldahl) assembly required for the distillation process 3. Rotary Evaporator required for removal of solvents
Consumables	10 lakh	Consumable cost for the proposed project will be used in glassware and chemicals
Man power	10 lakh	Technical persons is required with M.Sc. degree or equivalent degree to conduct the experiments, and analysis of the Product.
Rental/Incubation Services	2 lakh	Incubation charges, office space and Mentorship
IP	2 lakh	IP cost for Publish and Grant patent
Travel	1.5lakh	Travel cost for the proposed project will be used in scientific meetings, conferences and Technical workshops
External services	7 lakh	logo ,legal,, Consultancy , Compliances ,Testing ,Sample Characterization
Contingency	2.5 lakh	website, packing cost, bill book, Stationary , ,Exhibition ,Incidental Expenses etc.
Total (Max 50 Lakhs)	50 lakh	



Thank you